



Mark Dosen <mark@dosen1.com>

FW: Bipolar Ionization / Needlepoint Ionization for Indoor Air Quality Management

1 message

Ryan Schmit <RSchmit@thendesign.com>

Mon, Nov 30, 2020 at 9:00 AM

To: Mark Dosen <mark@dosen1.com>

Cc: Kathleen Mack <mackk@bbhcsd.org>, Ellen Kramer <kramere@bbhcsd.org>, Fred Pedersen <pedersenf@bbhcsd.org>, Brittany Wigman <wigmanb@bbhcsd.org>, Joelle Magyar <magyarj@bbhcsd.org>, Craig Yaniglos <yaniglosc@bbhcsd.org>, Christopher Coad <coadc@bbhcsd.org>

Good morning,

I hope everyone had a great Thanksgiving.

Please see the information below from Thorson Baker. Nate provides some history and information regarding the ozone concern. Also included is a narrative of why the Bipolar Ionization was the preferred choice for this project.

Let us know if you have any questions.

Thank you,

Ryan Schmit ra, leed ap, ncarb



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From: Nathaniel J. Kobb <NKobb@thorsonbaker.com>**Sent:** Monday, November 30, 2020 8:31 AM**To:** Ryan Schmit <RSchmit@thendesign.com>**Subject:** RE: Bipolar Ionization / Needlepoint Ionization for Indoor Air Quality Management

Ryan,

One of the primary concerns regarding Bi-polar ionization that was brought up was the creation of ozone. This was a very early concern with the technology when it came out over a decade ago but it is no longer a concern as the current technology produces very little ozone in concentrations that meet all indoor safety standards. Here is a quick link to the article showing how it passes UL requirements: <https://blog.plasma-air.com/plasma-air-needlepoint-bipolar-ionization-technology-passes-underwriters-laboratory-ul-2998-test-for-zero-ozone-emissions/>

Below is a very handy chart showing the pros/cons of Bi-polar ionization (Plasma Air in this chart) versus other air purification systems. TBA agrees that it is the most cost effective option for adding air purification to a school. Here is a link for independent lab studies on the effectiveness of the technology: <https://www.plasma-air.com/resources>

Here is a link to a school district that installed this recently to combat the pandemic: <https://blog.plasma-air.com/texas-school-district-installs-plasma-air-ionization-systems-to-help-protect-students-and-staff-from-airborne-viruses/>

HEPA is mentioned below but based on the equipment we have installed will not work due to the increased pressure drop and lack of space for the filtration.

UV lighting could be added at a substantial cost but according to our research is not as effective at killing viruses as quickly as the bi-polar ionization products.

Ventilation can be increased if the air is temperate but cannot be sustained throughout the year due to the system not being able to keep up, costs a lot in increased energy usage and potentially causes humidity and temperature issues by overwhelming the system.

Portable air cleaners with HEPA and/or UV can be installed but based on our research are extremely expensive and having one unit per room creates a maintenance issue.

How Plasma Air Compares to Reactive Air Purification Methods

Plasma Air's bipolar ionization technology is a superior solution because it proactively treats the air in the occupied space at the source of contamination. Traditional end-of-pipe solutions utilize a reactive "pass-through" or "filtered" approach.

	Plasma Air	Particle Filtration	HEPA / Fine Grain Filters ¹	Carbon Filters ²	Ultraviolet ³	Biofilters	Chemical Scrubber	PCO ⁴
Description	BIPOLAR IONIZATION	PRE-FILTERS, BAG FILTERS, PRE-TREATED FILTERS, FIBERGLASS FILTERS FOR LARGER AIRBORNE PARTICLES	CAPTURES FINE AIRBORNE PARTICLES	ABSORBS AND FILTERS CHEMICALS	ULTRAVIOLET LIGHT KILLS GERMS AND REMOVES AIRBORNE PARTICLES	TREATS EXHAUSTED AIR THROUGH BIOLOGICAL MEDIA BY REACTING WITH CONTAMINANTS	TREATS EXHAUSTED AIR THROUGH CHEMICAL MEDIA BY REACTING WITH CONTAMINANTS	FILTERED MEDIA IS COATED WITH MATERIAL THAT REACTS WITH CHEMICALS
Energy savings	UP TO 30%	NONE	NONE	NONE	NONE	NONE	NONE	NONE
Pressure drop	LOW	LOW	HIGH	MEDIUM	NONE	VERY HIGH	HIGH	MEDIUM
Particle size	SMALL	LARGE (> 5UM)	SMALL (< 0.01UM)	N/A	N/A	N/A	N/A	LARGE
Treats air in room	YES	NO	NO	NO	NO	NO	NO	NO
Treats Make-up / Supply air	YES	YES	YES	YES	YES	NO	NO	YES
Treats return air	YES	YES	YES	RARELY	YES	NO	NO	YES
Treats exhaust air	YES	RARELY	NO	YES	RARELY	YES	YES	RARELY
Capital costs	LOW	LOW	MEDIUM	HIGH	MEDIUM	VERY HIGH	VERY HIGH	HIGH
O&M Costs	LOW	LOW	LOW TO MEDIUM	HIGH	LOW	VERY HIGH	VERY HIGH	HIGH
Energy Costs	LOW	LOW	HIGH	HIGH	HIGH	HIGH	HIGH	MEDIUM
Disposal Costs	NONE	LOW	DEPENDS ON CONTAMINATION	HIGH	MEDIUM	HIGH	HIGH	HIGH
Success rates ⁵	99%	LOW	8-99.7%	LOW	LOW	LOW	LOW	LOW

1) HEPA filters add significant pressure drop resulting in larger horsepower motors and increased fan energy. 2) Carbon has high first cost, additional space requirements in the air handling units and prohibitive media replacement costs. 3) At high levels, UV can create noxious gases and is mutagenic. Bacteria kill rates are negligible with no effectiveness in the occupied space. 4) PCO can produce formaldehyde. 5) Success Rate based on independent tests on airborne pathogens

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From: Mark Dosen <mark@dosen1.com>
Sent: Monday, November 23, 2020 7:28 AM
To: Ryan Schmit <RSchmit@thendesign.com>
Cc: Kathleen Mack <mackk@bbhcsd.org>; Ellen Kramer <kramere@bbhcsd.org>; Fred Pedersen <pedersenf@bbhcsd.org>; Brittany Wigman <wigmanb@bbhcsd.org>; Joelle Magyar <magyarj@bbhcsd.org>; Craig Yaniglos <yaniglosc@bbhcsd.org>; Chris Coad <coadc@bbhcsd.org>; Mark Dosen <mark@dosen1.com>
Subject: Fwd: Bipolar Ionization / Needlepoint Ionization for Indoor Air Quality Management

Hi Ryan,

One of our parents sent us the information below on the effectiveness of bi-polar ionization. Please let us know if you have any additional thoughts or comments, no rush.

Thanks...Mark
----- Forwarded message -----

From: Giselle Kaschalk <gikalma@gmail.com><mailto:gikalma@gmail.com>>
Date: Fri, Nov 20, 2020 at 11:54 AM
Subject: Fwd: Bipolar Ionization / Needlepoint Ionization for Indoor Air Quality Management
To: Joelle Magyar <Magyarj@bbhcsd.org><mailto:Magyarj@bbhcsd.org>>, <Dosenm@bbhcsd.org><mailto:Dosenm@bbhcsd.org>>, Kathleen Mack <Mackk@bbhcsd.org><mailto:Mackk@bbhcsd.org>>, Fred Pedersen <Pedersenf@bbhcsd.org><mailto:Pedersenf@bbhcsd.org>>, Brittany Wigman <Wigmanb@bbhcsd.org><mailto:Wigmanb@bbhcsd.org>>

Board Members and Superintendent,

I want to address the agenda item from the November 18th Board Meeting: PreK-5 Ventilation Upgrade Discussion.
Mrs. Superintendent, thank you for being passionate about the new building and trying to get the best for our children. I want to thank the Board for making this process transparent to the district's parents.

After watching the last couple of meetings where you discussed the ventilation topic, I was curious to understand more about the technology proposed in the Ventilation Upgrade.

11/30/2020

Dosen1.com Mail - FW: Bipolar Ionization / Needlepoint Ionization for Indoor Air Quality Management

I reached out to a friend who does indoor air quality consulting. I asked for her professional opinion regarding the air cleaning technology proposed to be installed in the new elementary school. Please keep in mind that she is not familiar with the details of new elementary school building project. The information she knows is what I shared with her: the links posted on the board meeting agenda.

Based on the info I shared with her, it is her opinion that the proposed technology is not the best alternative since it can produce ozone, and there are not enough studies showing it is effective. As a parent, I was disappointed to learn this information. I, like many others, keep searching for ways to keep my child safe and protect.

I forward here the full content of her e-mail, which contains her professional assessment and credentials and other links she provided regarding this topic. My friend also made herself available if the Board and Superintendent have any further questions.

Thank you for your time,

Giselle Kaschalk

----- Forwarded message -----

From: Susann Geithner <sgeithner@geithnerconsulting.com<mailto:sgeithner@geithnerconsulting.com>>

Date: Thu, Nov 19, 2020 at 2:10 PM

Subject: Bipolar Ionization / Needlepoint Ionization for Indoor Air Quality Management

To: Gikalma@gmail.com<mailto:Gikalma@gmail.com> <Gikalma@gmail.com<mailto:Gikalma@gmail.com>>

Hi Giselle,

So the technology referenced in the links you provided is known as bipolar ionization / needlepoint ionization.

I want to start off by sharing ASHRAE's opinion (American Society of Heating Refrigeration and Air Conditioning Engineers) and the CDC. ASHRAE is responsible for the building code with regards to HVAC, electric and plumbing. Its ASHRAE 90.1/62.1 standard is referenced in all building codes throughout the states. Ohio building code references ASHRAE 90.1-2010 and ASHRAE 62.1-2010. They effectively write the code for air conditioning systems.

ASHRAE is a large and reputable association. They will therefore choose their words carefully and not endorse something unless proven to work. They also generally don't condemn a technology unless proven not to work. Please keep this in mind when reading their statements below.

As for myself, I'm an independent consultant, not affiliated with any engineering company or any company that sells HVAC system. For many years now my job has been to improve buildings. Indoor Air Quality (IAQ) is one of the topics I consult on and as a green building certification consultant my work includes checking the engineers work and consulting to improve IAQ. This means I review system design and calculations for outside air requirements per code, measure IAQ in new and existing buildings, consult to improve IAQ through added filtration, increase outdoor air, material selection to minimize off-gassing. I work on mostly commercial, government and institutional projects national and international including the LEED certification of the Cleveland Children's Museum. Since Covid came in our lives I have been following the medical and IAQ science and studies here in the states as well as in Germany and Switzerland. I'm also a LEED AP (accredited by the US Green Building Council) and a WELL AP (building professional for Health and Wellness in buildings) <https://www.wellcertified.com/> If anyone has any questions with regards to my professional experience, expertise, credentials or conflict of interest, feel free to ask.

My evaluation regarding this technology is that the technology has not been studied enough to show effectiveness. Both ASHRAE and CDC caution the use of this technology and point out that it produces ozone, which is an irritant causing breathing problems. Any equipment therefore needs to show that ozone production is below a safe level (50 or better 10ppb per latest research) or filtered out (typically with carbon filters). ASHRAE Position Document on Filtration and Air Cleaning <<https://www.ashrae.org/file%20library/about/position%20documents/filtration-and-air-cleaning-pd.pdf>> also notes that this technology is not recommended without the use of appropriate filtration. The reason being is that this technology works, I'm simplifying here, by charging the air to make particles clump together and therefore getting bigger and heavier and either falling to the ground or being filtered out by HVAC filter in the air ducts. The more powerful the ionizer the more ozone is produced. Also some studies for EPS (electrostatic precipitators; not the same but similar to what's proposed here) have shown silicon build up on the device, which led to a quick decay in effectiveness, which was only prevented by cleaning them every 2 months. See attached study of effectiveness of home air Cleaner sponsored by Health Canada, 2008.

General notes

At the end of this email you'll find links to resources and studies with regards to aerosols in indoor environments, air cleaning devices, studies in schools with regards to aerosol and my article. My articles go into more detail regarding Covid-19 and IAQ, filtration and many links to studies and resources.

Recommendation for a school setting

Increase outdoor air: Dilute and remove the pathogen. There is agreement in Germany and the US that CO2 concentration in indoor spaces is a great indicator for the amount of aerosols in a space. There is also increasing evidence that the doses of the virus is important in the severity of the disease. So the more fresh air is brought into a space that less CO2 and more mixing is taken place. For HVAC system this means increasing the amount of outside air being brought into a space and/or CO2 sensors, which regulate the outside air supply. So high CO2, due to higher occupancy will trigger increased outside air supply through the ventilation system. This however also increases utility cost, because more air is heated or cooled. The use of CO2 sensors, which control the supply of fresh air, can help reduce/optimize those operational cost. This is demand controlled ventilation, which is a common energy efficiency measure and also improves IAQ.

Natural or mixed mode ventilation: Being able to open the windows to increase fresh air supply is not always possible. Schools often don't have operable windows or outside noise will make it impractical. Designing schools to allow for natural ventilation in conjunction with mechanical ventilation will save operational cost and allows for more fresh air supply, which helps decrease aerosol concentration indoors.

Air filtration: The use of air filters in HVAC system is common place and not a new technology. New buildings now routinely use MERV 13 filter. However for filters to be effective at filtering out 99% of particle the size of aerosols they should be HEPA filter (MERV 17 or higher). A MERV 13 filter's effectiveness at filtering particles of 0.3-1.0 (size of most aerosols) is only 50%. See ASHRAE link for details. A higher MERV rating of filters means the HVAC fans have a harder time moving the air through the filter. So there is a pressure built-up and pressure drop associated with it. Fans need to be sized for this added resistance and the system needs to be tight. So the air can not evade the filter and find a way around it. Larger/more powerful fans means more energy consumption. How effective the use of filters is at removing aerosols also depends on the ratio of fresh air versus return (recirculated) air. More fresh air and more and better filtration provides indoor air with less aerosols. However per the Technical University of Berlin more fresh air is better than more filtered air, unless you have a HEPA filter installed.

UVGI, UV-C: This is recommended only in combination with filters. In duct installation with powerful enough and long enough equipment is preferred. The air needs to be exposed to the UV light for a long enough time to be effective. Maintenance is an issue, since the effectiveness decrease with the lamps getting dirty. If using UV-C Upper Air Disinfection (UV-lights mounted to the ceiling) reflection of those UV rays onto people is an issue. See more info at the ASHRAE link.

Portable Air Cleaner / Air Purifier: Almost all air purifier use a filter and often a combination of UV-C, ionization, electrostatic precipitators and carbon filter. There are 2 studies in Germany which looked at the effectiveness of air purifier at removing aerosols. One is actually done in a classroom the other in a room the size of a classroom. Both showed even without any ventilation the filters are effectively making a difference. One study estimated that within 1 hour of occupancy the room with air purifiers will have only 1 infection versus 3.5 in a room without. I included a summary of key findings of those 2 studies at the end and attached one of the studies, which is published in English.

Humidity and temperature: Optimal humidity and temperature has always been linked to better health of building occupants. New studies also show links to this being relevant for Covid infections. The science is pretty simple here. If your nose membranes aren't dried out due to low humidity, they are also more resistant to infections. See link below to the study.

<https://www.ashrae.org/technical-resources/filtration-disinfection#bipolar>

Bipolar Ionization/Corona Discharge / Needlepoint Ionization and Other Ion or Reactive Oxygen Air Cleaners

* Air cleaners using reactive ions and/or reactive oxygen species (ROS) have become prevalent during the COVID-19 pandemic. New devices that are not mentioned elsewhere in this guidance likely fall into this category.

* Technologies utilize various methods to create reactive ions in air that react with airborne contaminants, including viruses. The design of the systems can be modified to create mixtures of reactive oxygen species (ROS), ozone, hydroxyl radicals and superoxide anions.

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- * Systems are reported to range from ineffective to very effective in reducing airborne particulates and acute health symptoms.
- * Convincing scientifically-rigorous, peer-reviewed studies do not currently exist on this emerging technology; manufacturer data should be carefully considered.
- * Systems may emit ozone, some at high levels. Manufacturers are likely to have ozone generation test data.

For more information, see the ASHRAE Position Document on Filtration and Air Cleaning<<https://www.ashrae.org/file%20library/about/position%20documents/filtration-and-air-cleaning-pd.pdf>> and CDC Response to ASHRAE ETF on Bipolar Ionization<<https://www.ashrae.org/technical-resources/filtration-disinfection#cdcposition>>

CDC Position on Bipolar Ionization

ASHRAE does not currently have a Society position on bipolar ionization. However, the ASHRAE ETF did reach out to CDC for their position on the technology. The following is the response from CDC in its entirety:

Thank you for your question. Although this was pointed out in the earlier CDC responses, it is important for me to re-emphasize that CDC does not provide recommendations for, or against, any manufacturer or manufacturer's product. While bi-polar ionization has been around for decades, the technology has matured and many of the earlier potential safety concerns are reportedly now resolved. If you are considering the acquisition of bi-polar ionization equipment, you will want to be sure that the equipment meets UL 2998 standard certification (Environmental Claim Validation Procedure (ECVP) for Zero Ozone Emissions from Air Cleaners) which is intended to validate that no harmful levels of ozone are produced. Relative to many other air cleaning or disinfection technologies, needlepoint bi-polar ionization has a less-documented track record in regards to cleaning/disinfecting large and fast volumes of moving air within heating, ventilation, and air conditioning (HVAC) systems. This is not to imply that the technology doesn't work as advertised, only that in the absence of an established body of evidence reflecting proven efficacy under as-used conditions, the technology is still considered by many to be an "emerging technology". As with all emerging technologies, consumers are encouraged to exercise caution and to do their homework. Consumers should research the technology, attempting to match any specific claims against the consumer's intended use. Consumers should request efficacy performance data that quantitatively demonstrates a clear protective benefit under conditions consistent with those for which the consumer is intending to apply the technology. Preferably, the documented performance data under as-used conditions should be available from multiple sources, some of which should be independent, third party sources.

Testing mobile air purifiers in a school classroom: Reducing the airborne transmission risk for SARS-CoV-2 J. Curtiusa, M. Granzina, J. Schroda Institute for Atmospheric and Environmental Sciences, Goethe University Frankfurt am Main, Altenhöferallee 1, 60438 Frankfurt am Main, Germany <https://aktuelles.uni-frankfurt.de/forschung/studie-zeigt-luftreiniger-beseitigen-90-prozent-der-aerosole-in-schulklassen/>

Key take away:

- * Looked at classrooms, one with and one without air purifier.
- * Used air purifier with only HEPA filter, no ionizer or others added filtration or air treatment, units had CADR 333; 4 per room --> 5.5 air exchanges per hour
- * aerosol concentration decreased by more than 95% within 37 minutes following an exponential decay rate, while the room without air purifier remain at nearly constant levels of aerosols
- * estimated effect on transmission of the virus (modeled based on measured data of the aerosols in the room): after one hour the difference between new infections with and without air purifiers is a factor of 3.5 and it becomes a factor of 6.2 after 2 hours. So in a room with air purifier 1 person gets infected versus 3.5 people in a room without
- * to maintain a healthy level of CO2 in the room window ventilation is still required during breaks
- * the position of the air filter is important. Air intake and outlet shall not be obstructed by furniture
- * cold draft: air purifier blowing up instead of horizontal are preferred, so not to create uncomfortable drafts
- * regular maintenance needs to take place and filter inspected

Quantification of Viromed Klinik Akut V 500 Air purifier for the reduction of indirect COVID -19 infections via aerosols University of the German Army in Munich Institute of fluid mechanics and aero dynamics

The study is only in German, but shows how effective high quality air purifier can be. This particular unit costs \$ 4,000 compared to the unit used in the other study under \$100

- * assumes closed room, no mechanical ventilation
- * one or 2 air purifier (Viromed Klinik Akut V 500) providing 6 air changes per hour
- * this air purifier has 6 filters (pre filter, plasma generator, electrostatic filter, UV-C catalyst, microfilter (HEPA) + carbon filter, ionizer)
- * assumes people in the room don't wear masks
- * 50% reduction in aerosols after 12 minutes (2 air purifier) 25 minutes with one air purifier

<image001.png>

Aerosol concentration over one hour.

Left: class room; in black without air purifier, green one air purifier, red and blue 2 air purifier in different positions.

Right: corridor: black: without air purifier; green: long corridor with one air purifier; red: short corridor with one air purifier; blue: long corridor with 2 air purifier on opposite ends

Other resources:

https://blogs.tu-berlin.de/hri_sars-cov-2/blog/

<https://www.ashrae.org/technical-resources/resources>

<https://www.news-medical.net/news/20200924/Aerosol-based-study-of-COVID-19-risk-indoors-and-the-impact-of-preventive-measures.aspx>

<https://aaqr.org/articles/aaqr-20-06-covid-0302>

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11/30/2020

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<Indoor Air Quality in the age of Covid.docx> <EffectivenessofHomeAirCleanersreporttoEnvironmentCanada2008.pdf>

<2020.10.02.20205633v2.full.pdf>